# *Appendix B*
# *Fireground Tactics for*
# *Contamination Reduction*

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Operational Plan

## I. INTENT

Establishing Best Practices for exposure reduction during Fireground operations, we will reduce our members exposure onscene during all fire incidents by conducting onscene Personal Exposure Reduction (Post fire gross decon). In addition to gross decon, establishing hot, warm, and cold zones will reduce the opportunity for cross contamination between the hazard zones.

## II. PROCEDURE

These procedures will aid in reducing possible carcinogenic exposures before, during and after suppression activities.

### A. Fire incidents:

When personnel are exposed to environments that can potentially contaminate their clothing, equipment or self (skin, lungs, etc.) the presumption is that they have been contaminated. Some examples of hazardous environments or considerations are:

- Operations in environments where any amount of smoke is present.
- This includes ANY duration of time. The smell of products of combustion indicates and exposure. If you can smell the smoke, you are being exposed.
- Personnel should be cognizant that an exposure to products of combustion does not require visible smoke; inhalation and absorption of low doses, in the part per million (ppm) range or microscopic amounts, have a potential carcinogenic effect. Exposures are likely to occur during operations when in proximity to products of combustion such as:

    - Food on the stove (including when there is only smoke present)
    - Natural vegetation fires
    - Vehicle fires
    - Live fire training activities
    - Brush fires (trash, tires, fertilizers, pesticides, insecticides and unknowns).
    - Trash/Dumpster fires.
    - Driver/Engineer performing pumping operations.
    - Incident Commanders / Safety Officers performing Incident Command functions.
    - Un-deployed Rapid Intervention Crews.
    - Crews assigned to an exterior exposure line.
    - Unintended exposure of any personnel due to significant wind shift, scene dynamics or complications.
    - Fire Cause Investigations activities.
    - Ventilation operations.

**B. Personal Exposure Reduction Process: (Gross Decontamination)**

- As recommended NFPA 1500 (14.5.2) (2018 version): "The AHJ shall provide standard operation procedures related to the decontamination to the firefighters exposed to fire ground toxic contaminates".

- As recommended NFPA 1500 (14.5.3) (2018 version): "The AHJ should provide appropriate decontamination facilities and equipment for firefighters exposed to fireground toxic contaminates".

- Personnel Exposure Reduction, as a best practice, should be located upwind from the incident, and downwind from the rehabilitation/medical treatment area.

- Personnel decontaminating others should don appropriate PPE. (Nitrile gloves and an N95 mask at a minimum).

- Place a road cone to identify the Personnel Exposure Reduction area. Position a road cone with the garden hose with spray nozzle. The garden hose should be pumped with the pressure as low as possible.

- Place spray bottle of decon detergent (NFPA 1851 compliant) solution by the decon hose.

- Place the plastic brush next to the hose and decon detergent.

- Soft bristle scrub brushes and department-approved NFPA 1851 compliant soap/cleaner may be used to facilitate the cleaning process.

- It is a best practice that all fire personnel that were exposed to products of combustion be decontaminated prior to entering rehab or leaving the incident scene. After exiting the Hot Zone.
- It is recommended that crews remain on air, when possible, and report directly to the designated decon hose Line(s).

- Crews exiting the fire should be directed to the decontamination area.

**NOTE: As a best practice, members with the lowest air supply should be decontaminated first and as a rule, personnel are to remain on-air until gross decon is complete, again when possible.**

- Personnel should be careful to not saturate the inner lining of the PPE. The goal is to keep the PPE operationally dry on the interior, but rinsed as clean as possible on the exterior.

- Start with the front side of the firefighter (shoulders to legs) arms extended away from body at 90 degrees and feet shoulder width apart.

- Close all PPE pockets and open storm flaps exposing zipper.

- Maintaining crew integrity, they shall assist each other in rinsing off debris and products of combustion in a systematic and thorough manner from the collar-line down; being mindful of higher potential collection points such as the armpit and groin areas.

- Apply NFPA 1851 compliant cleaner on PPE.

- Scrub with brush to take off the ash, soot, and any products of the fire environment off of the PPE.

- Rinse with water from hose line. (If using a booster or engine hoseline do not use Class A foam).

- Turn around and complete the process on the back.

- Clean boots last.

- Rinse with water applied head to toe and front and back.

- Personnel may to go through a water only decon if only exiting for a fresh SCBA bottle and returning into the Hot Zone.

- SAFETY TIP: Be careful not to saturate or get water down the neck and soak the firefighter.

- Once gross decontamination is complete, PPE's can be "doffed" and placed in the "drop zone."

- The Drop Zone should be remote and downwind of rehab due to off-gassing PPE. Placement of tarps or salvage covers is suggested to designate this area.

- Personnel must resist the urge to consume food or hydrate until the gross decon process is completed and they have entered a clean rehab area.

- This procedure is not exclusive to crews that have been in the IDLH environment. This procedure may include Engineers who are pumping, crews manning safety lines or secondary lines, safety officers, RIC or Two Out, and Incident Commanders. Particulates can affect all who are on the fire scene.

- Fire Investigators will take the same steps as structural Firefighters to protect themselves from exposures to carcinogens, VOCs and PAH's through the course of the investigation.

- Fire Investigators are also exposed to airborne chemicals and particulates when performing an investigation.

**C. Doffing PPE (Cold/Drop Zone):**

- Once personnel have been decontaminated, they move from the decontamination area to the designated doffing area or "Drop Zone". It is recommended to position the following items at the Drop Zone:

- Any crew members assisting with SCBA bottle replacement, doffing of PPE gear, etc. shall wear latex gloves and an N95 mask on at a minimum.

- Nitrile gloves.

- Cleansing decon wipes. NFPA 1500 (14.5.3.1) (2018 version).

- Eye protection.

- Respiratory protection masks (N95 minimum).

- 42 to 60-gallon 6 mil plastic bags for contaminated PPE (2 per firefighter).

- Separate trash bag for soiled sanitation wipes and disposable towels.

- Road cone identifying the area as the "Drop Zone".

- Roll of duct tape to secure bags.

- Remove structure gloves avoid skin contact with exterior of glove, Use sanitation wipes to clean hands.

- Once PPE's have been doffed, each member should use cleansing decon wipes to wipe down face, neck, head, hands, arms, groin, and any areas where exposed skin may be affected. As recommended NFPA 1500 (14.5.3.1) (2018 version): Firefighter personal hygiene following exposure to fireground contaminates shall include the following: *Wiping skin areas near the interface of protective clothing and equipment with a wet wipe immediately after doffing.*

- It is recommended to have a hood/glove exchange in the drop zone.

**D. Returning to the incident:**

- Re-enter the Drop Zone and don your PPE.
- Crews will have to go through the Personal Exposure Reduction process once assignment is complete.

**E. Release from the Incident:**

- Ensure you have completed the Personal Exposure Reduction procedure.
    - Including using personal wipes on your Hands, face, neck/throat, and eyes.

- When possible, first arriving units involved in suppression activities should be released prior to others to start the decontamination process.

- All Contaminated PPE's should be bagged up and sealed with duct tape.

- As a best practice, all contaminated equipment should not be carried in the cab of any apparatus returning from a fire incident due to cross contamination.

- It is recommended to shower within the hour upon release from an incident. NFPA 1500 (14.5.3.1(2)) (2018 version).

- Clean the personnel first, then tools and equipment!

- It is recommended that resources should remain assigned to incident until all exposed personnel have showered and clean PPEs are placed on apparatus. This would be based on the individual agencies policies and procedures.

- It is recommended that contaminated PPE's that are bagged up, to be laundered per NFPA 1851, Standard on Selection, Care, and Maintenance of Protective Ensembles for Structural Fire Fighting and Proximity Fire Fighting. This is also based on the individual agencies policy and procedures. This should include all PPE including helmet (including the liner and chin strap), gloves, and boots.

- It is recommended when cleaning and decontaminating all equipment to wear nitrile gloves and N95 mask.

- Items that are to be cleaned at the station level are SCBA, radios (and radio straps if used), flashlights, helmets (outside and interior portion), it is recommended to put a new liner in if applicable), Thermal Imaging Camera, or any other equipment utilized that was exposed to the products of combustion. Approved cleaners should be used per individual agency policy and procedures as well as manufacturer recommendations.

**F. Best Practices during Overhaul:**

- Wear full personal protective equipment throughout the duration of incidents, including self-contained breathing apparatus on air during salvage and overhaul operations.

- It is recommended to wait for 30 minutes post knock down to begin overhaul operations.

- Firefighters assigned the tasks of overhaul will need to be decontaminated once overhaul is complete.